

Shashikant

Senior MLOps Engineer

SUMMARY

Software Engineer with 10+ years of overall experience, including 6+ years specializing in MLOps, building and operationalizing scalable machine learning systems across cloud environments. Strong expertise in Azure Databricks, PySpark, and MLflow, with hands-on experience migrating and optimizing ML workloads from GCP to Azure. Adept at designing robust CI/CD pipelines, improving model performance, and ensuring cost-efficient execution in distributed systems. Known for strong collaboration with data science and engineering teams, a deep understanding of data pipelines, and a proactive approach to debugging, monitoring, and continuous optimization in production environments.

TECHNICAL EXPERTISE

- **Programming & Backend Engineering:** Python, FastAPI, Pydantic, Backend Development, Scalable API Development
- **Data Engineering & Processing:** PySpark, Data Pipelines, Data Engineering, Distributed Data Processing
- **Machine Learning & MLOps:** MLOps, ML Model Deployment, Model Operationalization, Model Optimization, Model Versioning, Model Monitoring, ML Lifecycle Management
- **Databricks Ecosystem:** Azure Databricks, Databricks Workflows, Databricks Runtime, Databricks Optimization
- **Model Tracking & Experimentation:** MLflow, Experiment Tracking, Model Registry
- **Cloud Platforms:** GCP, Azure, Cross-Cloud Integration, Cloud Data Movement
- **CI/CD & Automation:** GitHub Actions, CI/CD Pipelines, Workflow Automation, Deployment Automation
- **Containerization & DevOps:** Docker, Containerized ML Services
- **Monitoring & Performance:** Logging, Monitoring, Performance Optimization, Debugging
- **Version Control & Collaboration:** Git, Agile Methodology, Cross-Functional Collaboration
- **Engineering Practices:** System Design, Scalable Architecture, Technical Documentation

PROFESSIONAL EXPERIENCE

Webiwork Technologies Pvt. Ltd. | Senior MLOps Engineer

Jan 2016 – Present | Indore

- Led migration of machine learning models from GCP to Azure Databricks, analyzing existing architectures and optimizing them for performance, scalability, and cost-efficiency in a distributed environment.
- Built and optimized scalable data and ML pipelines using Databricks and PySpark, leveraging execution plan analysis and tuning techniques to improve processing efficiency and reduce compute costs.

- Designed and implemented CI/CD pipelines using GitHub Actions, enabling automated model deployment, testing, and version control aligned with MLOps best practices.
- Managed end-to-end ML lifecycle using MLflow, including experiment tracking, model versioning, and deployment, ensuring reproducibility and streamlined operationalization.
- Collaborated with Data Science and Data Engineering teams to enable seamless cross-cloud data movement (GCP ↔ Azure), developing automated workflows for data transfer, monitoring, and reliable model execution in production.

PROJECTS

Project 1: Enterprise Knowledge Intelligence Platform (RAG + Conversational AI)

Role: MLOps Engineer / ML Platform Engineer

Project Description: Led the operationalization and cross-cloud migration of NLP-based RAG pipelines from GCP to Azure Databricks, enabling scalable, cost-efficient model inference and lifecycle management. Designed end-to-end ML pipelines for embedding generation, retrieval, and conversational inference with robust tracking, monitoring, and CI/CD orchestration.

Roles & Responsibilities:

- Migrated and optimized NLP pipelines to Azure Databricks using PySpark, improving compute efficiency by ~30% and ensuring scalable distributed processing for large document datasets.
- Implemented MLflow-based model lifecycle management with GitHub Actions CI/CD pipelines, enabling versioning, automated deployment, and reproducible experiments across GCP ↔ Azure environments.

Technologies & Tools Used: Python, PySpark, Azure Databricks, MLflow, GCP (GCS, BigQuery), Azure (ADLS), HuggingFace Transformers, Docker, GitHub Actions (CI/CD), REST APIs, PostgreSQL, Redis

Project 2: AI Customer Support Automation Platform (Voice + Text)

Role: MLOps Engineer

Project Description: Built and productionized a scalable conversational AI system, focusing on ML pipeline orchestration, model deployment, and cross-cloud data flow between GCP and Azure Databricks for real-time inference workloads.

Roles & Responsibilities:

- Designed and deployed end-to-end ML pipelines using Databricks + PySpark, enabling real-time voice/text inference and improving processing throughput by ~40%.
- Built CI/CD pipelines using GitHub Actions and integrated MLflow tracking, ensuring automated testing, deployment, and monitoring of ML models in distributed environments.

Technologies & Tools Used: Python, PySpark, Azure Databricks, MLflow, GCP, Azure, Docker, GitHub Actions, REST APIs, Redis, PostgreSQL

Project 3: AI-Powered Candidate Evaluation & Analytics Platform

Role: MLOps Engineer

Project Description: Operationalized multilingual NLP models by building scalable ML pipelines and automated deployment workflows, ensuring high-performance model execution and seamless integration with cross-cloud data systems.

Roles & Responsibilities:

- Developed scalable PySpark-based data and ML pipelines in Databricks, enabling distributed processing of high-volume candidate data and reducing processing time by ~35%.
- Implemented model versioning, testing, and monitoring using MLflow and CI/CD pipelines, ensuring consistent model performance and faster release cycles.

Technologies & Tools Used: Python, PySpark, Azure Databricks, MLflow, GCP, Azure, Docker, GitHub Actions, PostgreSQL, REST APIs

Project 4: AI Resume Screening & Ranking System

Role: ML Engineer / MLOps Engineer

Project Description: Designed and deployed a production-grade ML inference system, focusing on model serving, lifecycle management, and performance optimization using Databricks and cloud-native tooling.

Roles & Responsibilities:

- Deployed scalable **model inference pipelines on Azure Databricks**, optimizing execution plans and reducing latency by ~30% through PySpark optimizations.
- Built **CI/CD workflows and MLflow integration** for automated model deployment, testing, and monitoring across cloud environments.

Technologies & Tools Used: Python, PySpark, Azure Databricks, MLflow, Docker, GitHub Actions, PostgreSQL, REST APIs

Project 5: Real-Time Customer Sentiment & Feedback Intelligence Platform

Role: MLOps Engineer

Project Description: Engineered a real-time ML pipeline for sentiment analysis, focusing on streaming data processing, model monitoring, and cross-cloud data movement between GCP and Azure environments.

Roles & Responsibilities:

- Built scalable streaming and batch pipelines using PySpark on Databricks, enabling real-time sentiment analysis on high-volume datasets with improved throughput by ~40%.
- Implemented end-to-end model monitoring, logging, and CI/CD pipelines, ensuring reliability, observability, and performance optimization in production systems.

Technologies & Tools Used: Python, PySpark, Azure Databricks, MLflow, GCP, Azure, Docker, GitHub Actions, REST APIs, PostgreSQL, Redis

EDUCATION

Rajiv Gandhi Prodyogiki Vishwavidyalaya
Bachelor's degree, Computer Science
Jun 2009 – May 2013